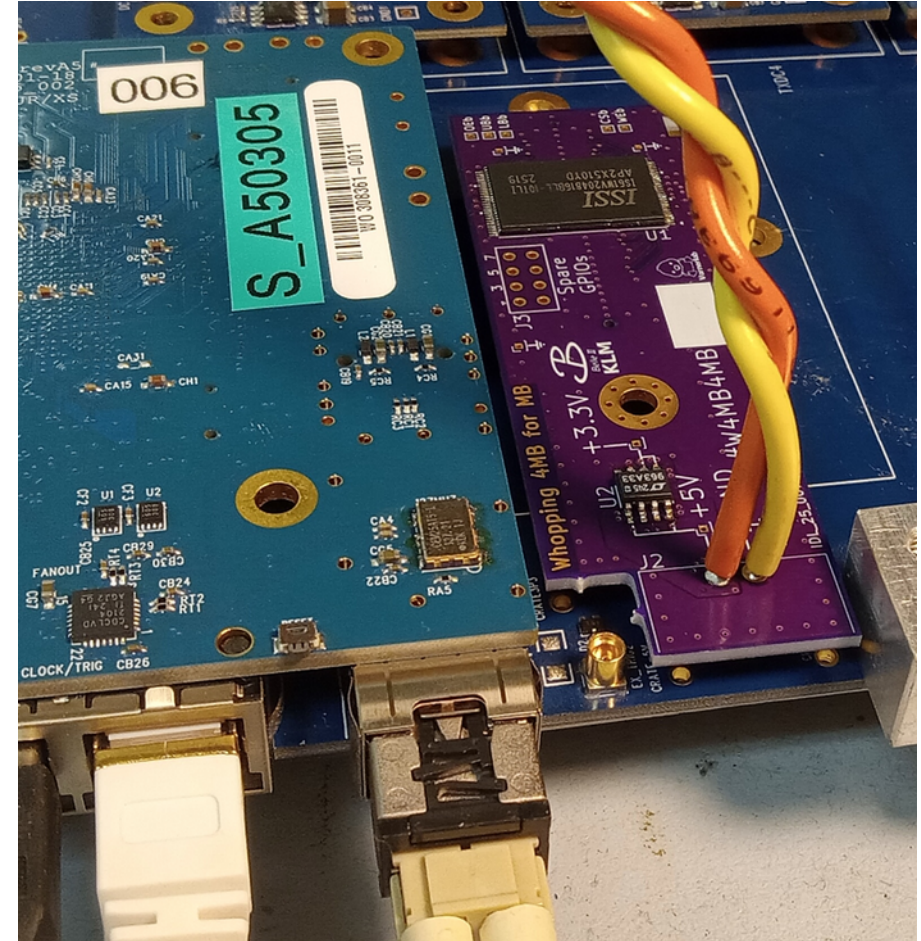


SRAM UPDATE

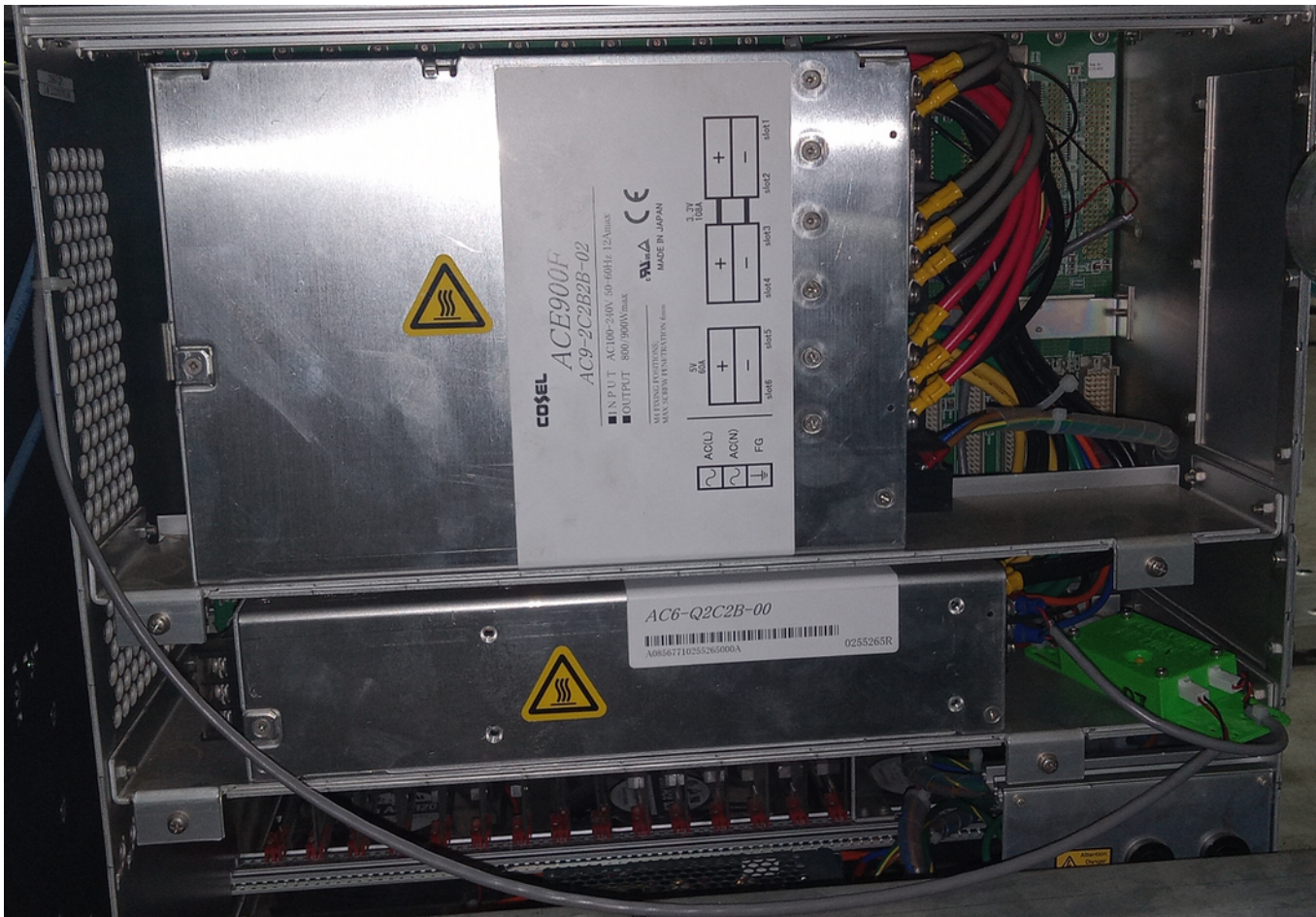
- ☕ Last time: Prototype PCB fully working
- ☕ These slides
 - 🌰 Condense information about requirements for rear transmission module (RTM) that supplies power and requirements for harness



BACK SIDE OF KLM COPPER CRATES

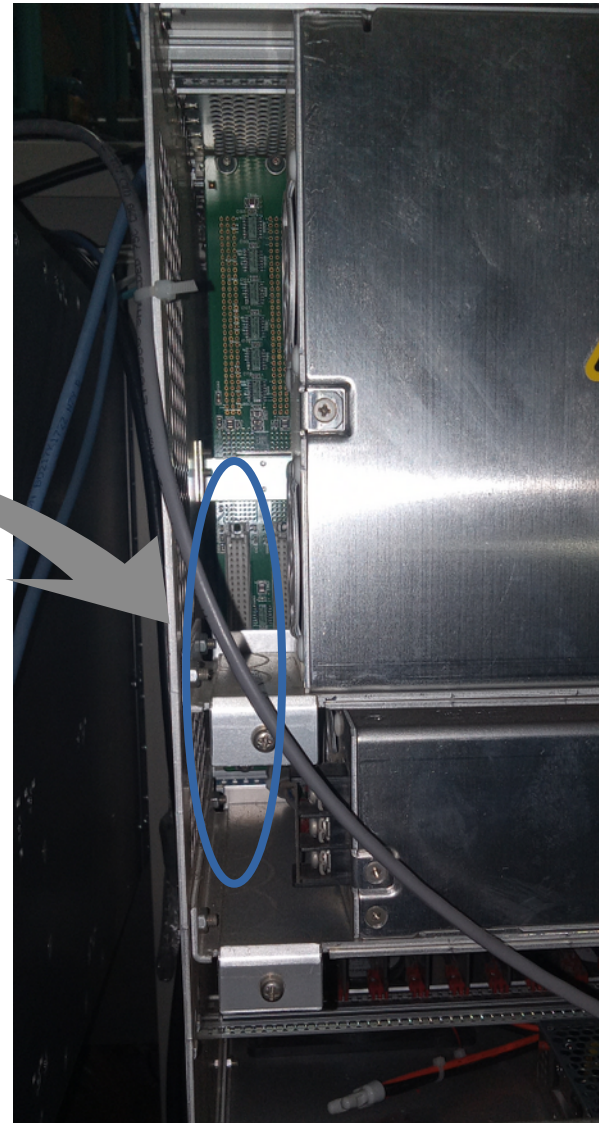
Upper power
supply

Lower power
supply



REAR POWER BOARD

- ☕ Board will go here
- ☕ Planned as a rear transition module (RTM)
 - ☕ Effectively, it's a board that plugs into the reverse side of a crate's backplane
- ☕ Lower power supply tray as deeper than that of the upper putting it closer to the planned location of the board

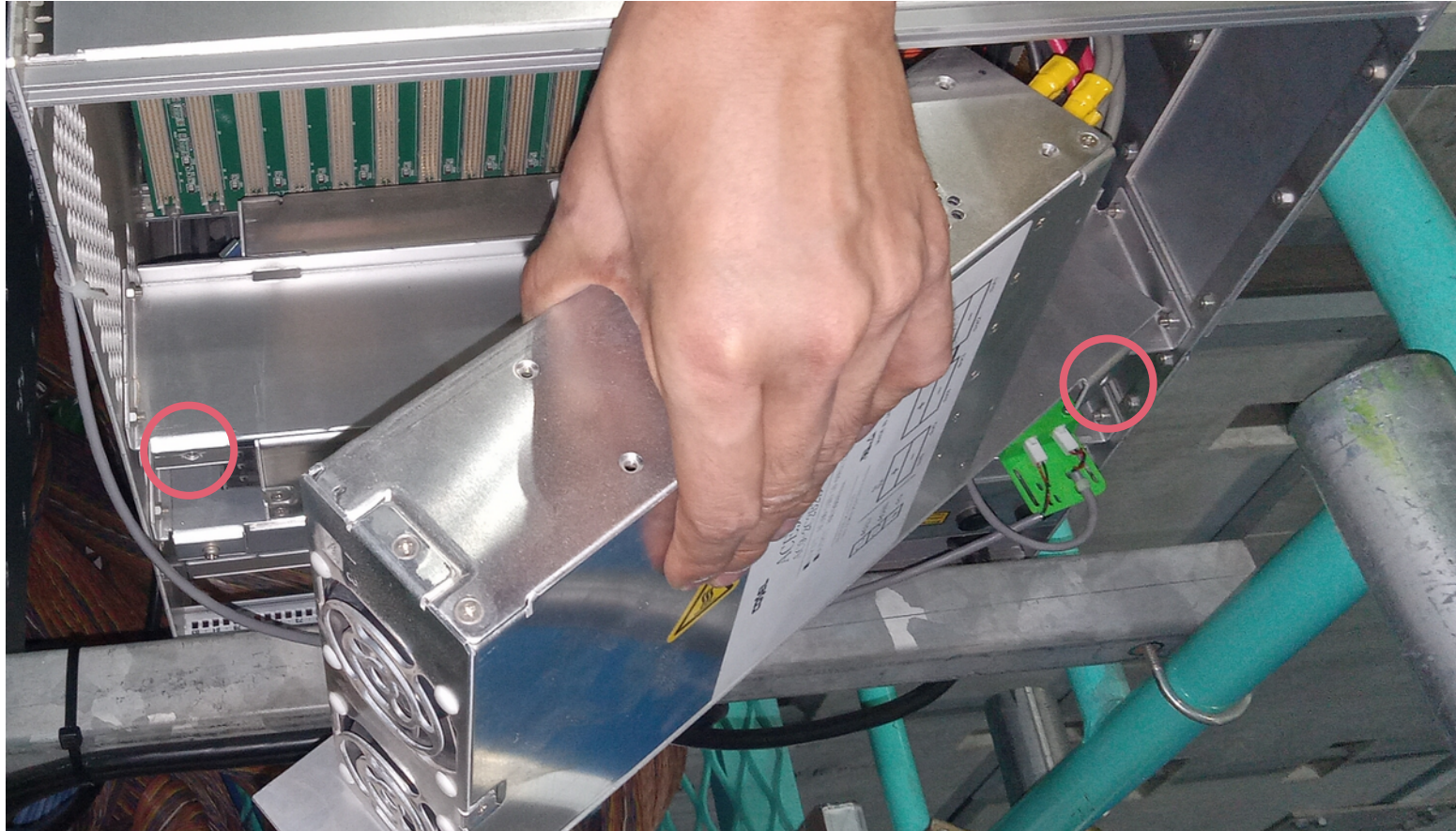


Upper power supply

Lower power supply

INSTALLATION EASE

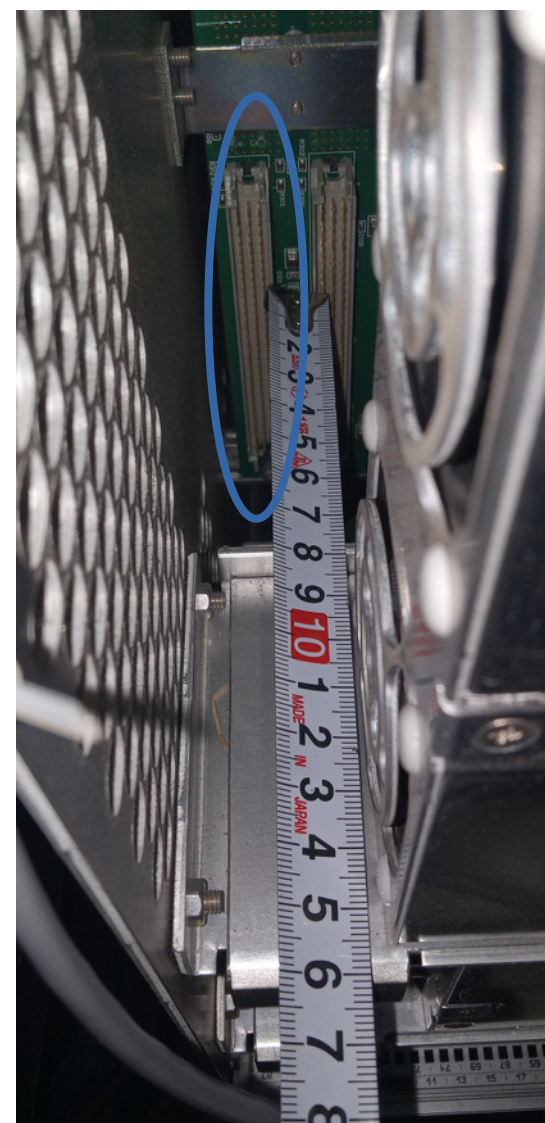
☕ Upper power supply easily pivots out of the way after removing **two** Phillips-head screws



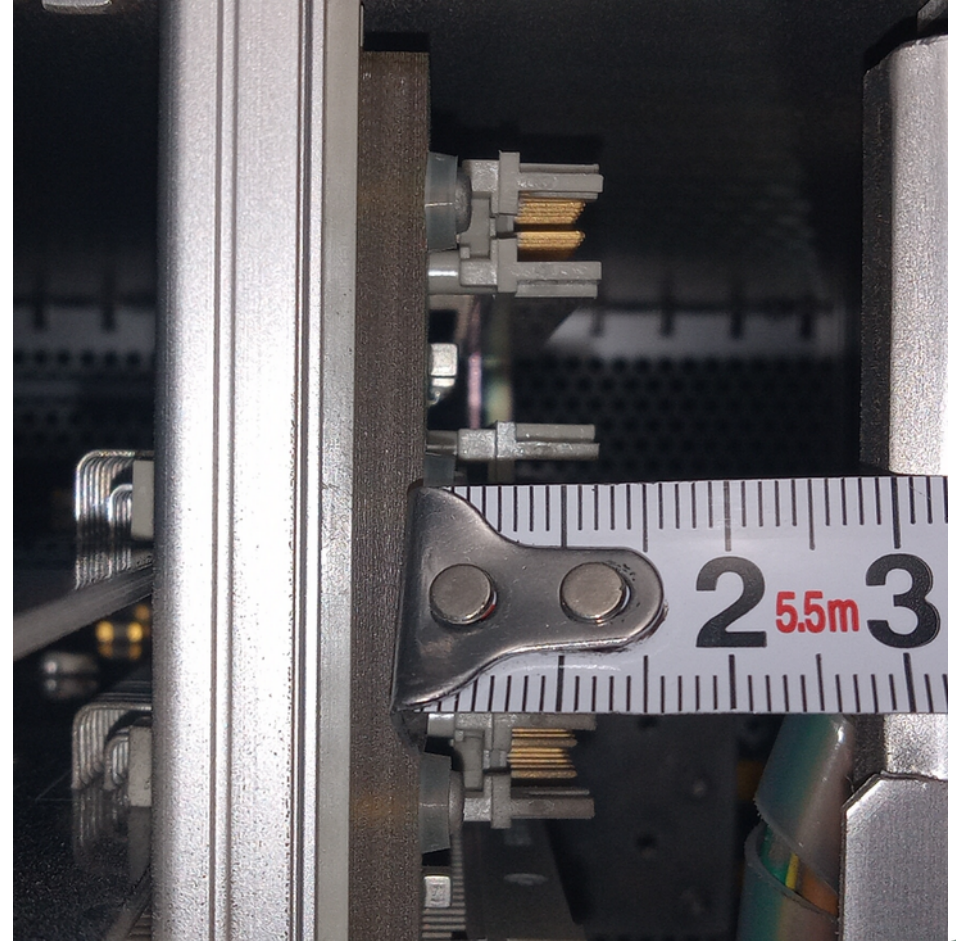
BOARD ORIENTATION / WIRE HARNESS ROUTE

☕ Board was envisioned as a right-angle 96-position VME backplane connector (male housing / female pins) and two plated through holes for soldering the +5 / GND wires of the wiring harness

☕ Tape measure shows the path through which the wiring harness will egress



- ☕ Envelope is 1.8 cm from the lower power supply tray to the edge of the connector (this is how much space we have to maneuver our board into place without removal of the metal tray that the lower power supply sits on)



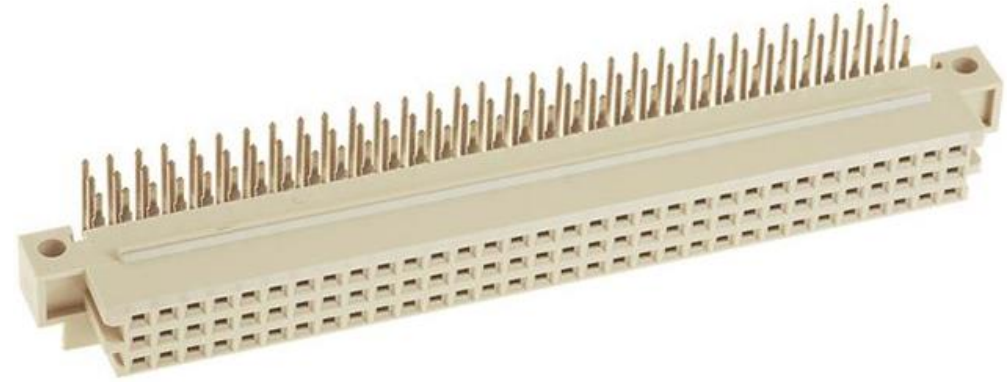
CONNECTOR

☕ Although board was envisioned as having a right-angle connector (1195-2893-ND), we actually have many of the straight version in stock (1195-1197-ND)

☕ This would also be possible, and perhaps simpler

☕ A 3rd option is to just connect ring terminals to the main output of the +5 V power supply

☕ I think we disfavored this because it's not proper --- those should connect the power supply to the bus bar and nothing else!



PINOUT

- ☕ +5 V is on pins 1, 13, and 32 of row B
- ☕ GND is on pins 2, 12, 22, and 31 of row B
- ☕ Every other pin should be left unconnected in the design of the rear power board

Pin Assignments for J2 Connectors					
(Pin rows in each connector are Z, A, B, C, and D)					
Pin Number	Row Z Signal Mnemonic	Row A Signal Mnemonic	Row B Signal Mnemonic	Row C Signal Mnemonic	Row D Signal Mnemonic
1	UD	UD	+5V	UD	UD
2	GND	UD	GND	UD	UD
3	UD	UD	RETRY*	UD	UD
4	GND	UD	A24	UD	UD
5	UD	UD	A25	UD	UD
6	GND	UD	A26	UD	UD
7	UD	UD	A27	UD	UD
8	GND	UD	A28	UD	UD
9	UD	UD	A29	UD	UD
10	GND	UD	A30	UD	UD
11	UD	UD	A31	UD	UD
12	GND	UD	GND	UD	UD
13	UD	UD	+5V	UD	UD
14	GND	UD	D16	UD	UD
15	UD	UD	D17	UD	UD
16	GND	UD	D18	UD	UD
17	UD	UD	D19	UD	UD
18	GND	UD	D20	UD	UD
19	UD	UD	D21	UD	UD
20	GND	UD	D22	UD	UD
21	UD	UD	D23	UD	UD
22	GND	UD	GND	UD	UD
23	UD	UD	D24	UD	UD
24	GND	UD	D25	UD	UD
25	UD	UD	D26	UD	UD
26	GND	UD	D27	UD	UD
27	UD	UD	D28	UD	UD
28	GND	UD	D29	UD	UD
29	UD	UD	D30	UD	UD
30	GND	UD	D31	UD	UD
31	UD	UD	GND	UD	GND
32	GND	UD	+5V	UD	VPC

CABLE SELECTION

☕ SRAM Expansion PCB prototype draws 16 mA on the testbench (TODO: recheck current during continuous operation)

☕ Accordingly, EKLM FWD crates will draw an additional 224 mA from the +5 V net

☕ Assuming safety factor of 4, current draw is 1 A

☿ 20 AWG copper wire is rated for 11 A and has resistance .01015 mOhms / ft

☿ At 5 ft harness length, voltage drop would be 50 mV (realistically 12 mV w/o safety factor)

☕ Smaller than 20 AWG may be too weak mechanically, e.g. if it were snagged

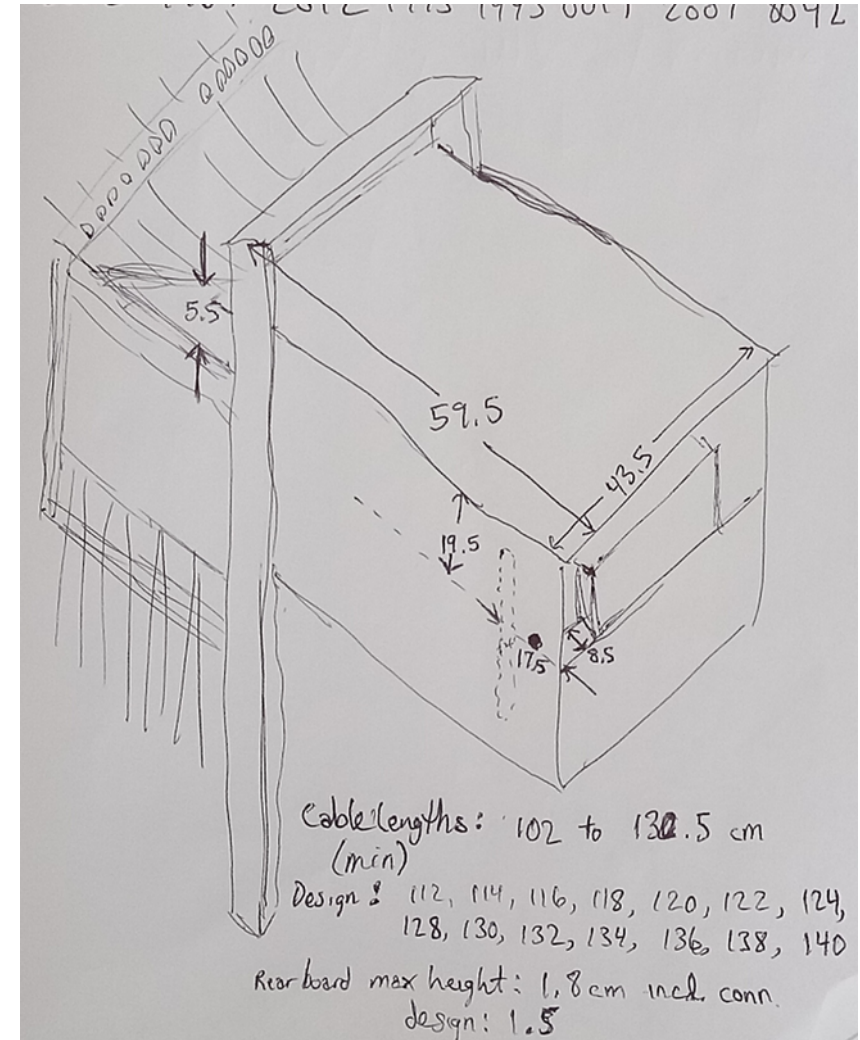
☕ Actually, I'd like 10 AWG or 8 AWG from the rear power board and branch out to 20 AWG at each of the 8 (barrel), 12 (EBWD) or 14 (EFWD) drops per harness

☿ 10 AWG is about 10 x the area of 20 AWG

☿ 8 AWG is about 16 x the area of 20 AWG

HARNESS SPECIFICATION

- ☕ Heavier gauge wire (e.g. 10 AWG) should be 95 cm in length
 - 🍎 90° bends should be formed at 16.5 cm, 31 cm, and 90.5 cm
 - 🍎 The last two should have protection against chaffing against the crate
- ☕ Drop wires (i.e. 20 AWG) should be in lengths of 12, 14, 16, 18, 20, 22, 24, 28, 30, 32, 34, 36, 38, and 40 cm
 - 🍎 Note the 4 cm gap between the 7th and 8th
- ☕ For the four crates in the barrel, the lengths are yet to be determined
 - 🍎 Probably something like 14, 16, 20, 22, 28, 30, 36, 38 cm



ADDITIONAL CONSIDERATIONS



Probably we want a fuse



- either a polyfuse or a serviceable one-time fuse

- if one-time fuse, we can maybe use an inline fuse



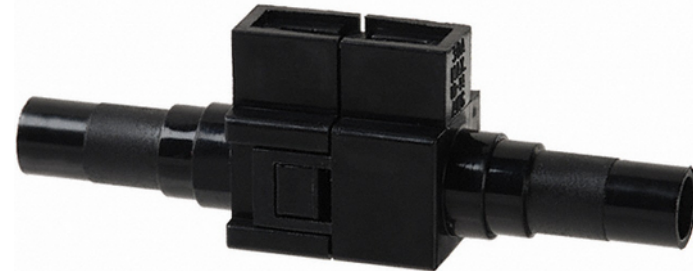
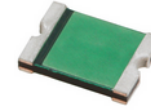
- 1.5x the operating current is typical (rounded up to next available size)



- I like 4x actually --- large enough to never have problems yet small enough to prevent fires



Further, our choice of wire and sheathing should meet KEK fire code standards
(How do we even find this information?)



SRAM PCB CONNECTOR

- ☕ Quoted connector may interfere with nearby aluminum bracket
- ☕ Difficult to find slim 2-position connectors, plus, if I order something weird, we may not have the right crimp tool
- ☕ Chosen connector must have a latching system
 - 🌰 Latch should be on top to avoid nearby aluminum bracket
 - 🌰 Does not have to be super slim if we extend SRAM PCB to extend past edge of motherboard
 - Connector can then be on bottom (facing motherboard) so it does not overlap with adjacent motherboard

☕ Now considering Molex 4-position connectors (WM4302-ND, WM2002-ND)

- 🌰 We use Molex quite extensively and have most of the tools
- 🌰 4 pins provide 2x redundancy
- 🌰 \$0.45 per pair (\$0.30 / \$0.15)
- 🌰 Friction lock

